Coral Reefs (2019) 38:45–61
https://doi.org/10.1007/s00338-018-01754-9

REPORT

Differential protein abundance during the first month of regeneration of the Caribbean star coral *Montastraea cavernosa*

Ryan A. Horricks[1,2] · Christophe M. Herbinger[3] · Brandon N. Lillie[1] · Paul Taylor[4] · John S. Lumsden[1,2]

Received: 10 September 2018 / Accepted: 13 November 2018 / Published online: 20 November 2018
© Springer-Verlag GmbH Germany, part of Springer Nature 2018

Abstract It is critical to determine the methods by which coral colonies regenerate tissue lost to physical injury as they provide the physical structure of coral reef systems. To explore regeneration, circular lesions (12 mm diameter × 3 mm depth) were created in the fall of 2014 on 124 *Montastraea cavernosa* colonies located in the coastal waters of Grenada and Carriacou (10–12 m depth). Coral regeneration was documented at weekly intervals for 28 days. Repeated measures ANOVA on estimated weekly coral regeneration rates showed that island ($p = 0.024$) and colony colour ($p = 0.024$) were the only factors significantly affecting lesion regeneration. Mean rate of lesion closure during the first 28 days was approximately 2.8 $mm^2\ d^{-1}$. Four identical circular lesions were created on 30 *M. cavernosa* colonies (Carriacou, 10–12 m depth) in the fall of 2015. One representative lesion created on each coral colony was re-sampled at each of 14, 21, and 32 or 33 days following injury, and coral tissue was flash-frozen. Tissues from 10 normally pigmented brown colonies were selected for proteomic analysis using tandem mass tags. The initial polyp sample, the day 14, and the final samples were used to quantify the difference in protein abundance as the lesions healed. In the tissue samples 6419 peptides were reliably identified, which corresponded to 906 unique proteins. During the first month of regeneration, 111 proteins were differentially abundant ($p < 0.05$) on at least one timepoint and of these, 11 were associated with regeneration. An additional 14 proteins were also identified that were differentially abundant ($p < 0.05$) and were associated with inflammation or antioxidant activity. This work demonstrates, for the first time, the differential abundance of proteins associated with regeneration in a scleractinian coral.

Keywords Regeneration · Proteomics · Tandem mass tag · *Montastraea cavernosa* · Grenada

Topic Editor Morgan S. Pratchett

Electronic supplementary material The online version of this article (https://doi.org/10.1007/s00338-018-01754-9) contains supplementary material, which is available to authorized users.

✉ Ryan A. Horricks
horrickr@uoguelph.ca

[1] Department of Pathobiology, Ontario Veterinary College, University of Guelph, Guelph, ON N1G 2W1, Canada

[2] Department of Pathobiology, School of Veterinary Medicine, St. George's University, True Blue, Grenada

[3] Department of Biology, Dalhousie University, Halifax, NS B3H 4R2, Canada

[4] SPARC BioCentre, The Hospital for Sick Children, Toronto, ON M5G 0A4, Canada

Introduction

Montastraea cavernosa is one of the remaining commonly encountered reef-building species of scleractinian coral in the Caribbean. As a large, sessile organism, it is susceptible to various forms of physical injury including predation, destructive fishing practices, boat and anchor grounding, careless SCUBA divers, sedimentation, storms, and disease (Lesser 2004). Any physical damage to coral tissue can follow one of two routes. Either the wound fails to heal and leaves exposed skeleton, or the whole structure re-grows completely. The complete healing of lesions and return to normal tissue function is referred to as regeneration (Goss 1992), and it plays an important role in coral survival (Foster et al. 2007). Coral colonies that cannot regenerate

tissue efficiently are susceptible to invasion from both encrusting and boring organisms, which may lead to the eventual death of the colony (Meesters et al. 1994). As scleractinian corals provide the three-dimensional structure of coral reefs (Squires 1958), their ability to regenerate tissue is of critical importance to the long-term health of the reef ecosystem as a whole.

Tissue regeneration rates vary between all species of hard coral, but the events leading to regeneration follow the same general pattern (Bak and Steward-Van Es 1980; Hall 1997; Marshall 2000; Meesters and Bak 1993; Meesters et al. 1996; Meesters et al. 1997; Nagelkerken and Bak 1998; Van Veghel and Bak 1994). Regeneration in the closely related *Orbicella annularis* (formerly *Montastraea annularis*) begins with the polyps surrounding a lesion producing an undifferentiated tissue layer that spreads inwards (Fisher et al. 2007; Meesters et al. 1994). New polyps appear in this tissue within approximately 2 weeks of lesion creation and begin to secrete septa (Fisher et al. 2007; Meesters et al. 1994). Regenerating polyps are initially semi-transparent but attain normal tissue pigmentation by the end of the regenerative process (Work and Aeby 2010). A mathematical model describing lesion closure has been proposed (Meesters et al. 1997) and modified (Fisher et al. 2007) that takes the following form:

$$y = y_o + \left[a\mathrm{e}^{(-b\times d)}\right]$$

where y is the current size of the lesion, y_o is the asymptote in the event that a lesion does not completely heal, a is the surface area of tissue that has been regenerated, b is the slope of the curve, and d is the number of days post-lesion creation.

Both regeneration and incomplete wound healing follow a similar sequence of events including coagulation, inflammation, proliferation, and finally, to some degree, tissue reorganization. In scleractinian corals, lesion regeneration has been described several times in multiple taxa using both gross and histological methods (Galloway et al. 2007; Palmer et al. 2011; Renegar 2015; Rodríguez-Villalobos et al. 2016; Work and Aeby 2010). Within the first 6 h following tissue damage, coagulation occurs with a clot comprised of extracellular aggregates (Palmer et al. 2011). In *M. cavernosa*, the inflammatory phase of tissue repair is facilitated by granular amoebocytes that fuse to become new tissues at the site of the lesion (Renegar 2015). Degranulation of granular cells expels cytotoxic material which functions as part of the innate immune response (Palmer et al. 2011). These amoebocytes arrange themselves in an epithelial-like layer by spreading around the lesion (Palmer et al. 2011). Granular cells, hyaline cells, and fibroblasts are all observed surrounding a regenerating lesion (Palmer et al. 2011) and have been reported in cnidarians as well as several other invertebrate groups (Palmer et al. 2011; Soderhall and Smith 1986; Wootton and Pipe 2003; Young 1974). Hyaline cells have been suggested to produce fibronectin and collagen, thereby facilitating the production and reformation of the extracellular matrix (Palmer et al. 2011; Soderhall and Smith 1986; Wootton and Pipe 2003; Young 1974). Following the inflammatory phase, cellular proliferation dominates (Palmer et al. 2011; Work and Aeby 2010), characterized by a thickening of the epithelium. Proliferation, along with reorganization, continues until the wound is completely healed and has regained its original colouration via uptake of zooxanthellae (Palmer et al. 2011; Work and Aeby 2010).

The mechanisms by which these processes occur are well studied in vertebrates and some 'model' invertebrates such as *Hydra* spp. (Technau and Steele 2011; Wenger et al. 2014). Studies on scleractinian corals are more limited, and to our knowledge, no information is available at the proteomic level. RNA sequencing (RNA-seq) experiments have been used to examine transcriptomic changes in *Hydra* spp. and the majority of identified pathways function in a manner analogous to those in vertebrates (Wenger et al. 2014). The few similar experiments conducted in coral species have found transcriptomic evidence for the upregulation of signalling pathways of several immune-related genes during tissue regeneration (Fuess et al. 2016; van De Water et al. 2015a, b). Transcripts related to the immune processes of pathogen recognition, signalling, and the resultant effector response have also been identified in scleractinian corals (Fuess et al. 2016). Following exposure to a pathogen-associated molecular pattern from *Escherichia coli*, seventeen transcripts that could be linked to immune-related processes in the scleractinian coral *O. annularis* were identified (Fuess et al. 2016). The immune response and regenerative processes are tightly linked and, at some level, indistinguishable from each other (Aurora and Olson 2015). While genomics and RNA-seq provide information as to how these processes might be initiated, proteomics can, within the limits of sensitivity, provide information as to which proteins are significantly altered (Slattery et al. 2012). Gaining insight to the presence of the effector proteins involved in these processes is of critical importance to further our knowledge of regeneration in a basal metazoan.

The purpose of the present study was to test the hypothesis that lesion size of *M. cavernosa* colonies would decrease exponentially with time up to an asymptote determined by the initial lesion size (proposed by Fisher et al. 2007; Meesters et al. 1997). The reparative abilities of *M. cavernosa* colonies were compared on patch reefs between two Grenadian islands. Once appropriate statistical models had been developed to determine sampling

times, proteomic analysis of coral tissue was used to determine which proteins were differentially abundant (DA) in *M. cavernosa* during the first month of the regenerative process.

Methods

Study sites

Four previously established study sites were selected to complement an existing dataset from Grenada (Anderson et al. 2012, 2014; Table 1) and four visually similar sites were also established on the nearby island of Carriacou (Table 1). Data on lesion regeneration were collected from late September to November 2014 and from November to December 2015.

Lesion regeneration

Circular lesions were created on 124 *M. cavernosa* colonies in the nearshore waters of Carriacou in September and Grenada in October 2014 at established benthic monitoring sites (Table 1). Identical lesions were created on 30 *M. cavernosa* colonies in the nearshore waters of Carriacou in October 2015 at two established benthic monitoring sites, Sandy Island and Jackadam (Table 1) (Grenada Ministry of Forestry, Fisheries & Agriculture and St. George's University, permit number IACUC 14001).

Lesions were created on *M. cavernosa* colonies to remove the tissue and a small amount of skeleton of a single polyp. Polyp core samples were collected using a 12.7 mm diameter diamond coring drill bit attached to a Stanley FatMax pneumatic drill (Stanley Black & Decker, New Britain, CT) powered by a dedicated SCUBA air cylinder. Coral lesion cores were drilled 3 mm ± 1 mm (average ± SE) deep in each colony. At the start of the experiment, lesions were always surrounded by live, normally pigmented tissue of hemispherical colonies of approximately the same size (0.097 m^3 ± 0.127, average ± SE). Each lesion was centred on a single polyp with the intention of removing a whole polyp and causing minimal incidental damage to neighbouring polyps. All coral colonies sampled were 10.20 ± 1.37 m (average ± SE) below the surface of the water. Upon removal from the colony, polyps were placed in individual pre-labelled sample collection tubes in ambient seawater. Once samples were brought to the surface, they were immediately rinsed three times in deionized water before being placed in individually labelled 1.5-ml centrifuge tubes and transferred to a MVE SC 20/12 V dry shipper (Core Cryolab, Toronto, ON) for flash-freezing in liquid nitrogen at − 120 °C. Polyps were stored in the field in the MVE SC 20/12 V dry shipper (Core Cryolab) and were transferred to a − 80 °C freezer upon arrival at the University of Guelph.

Lesions were photographed after lesion creation at day 0 and then at days 7, 14, 21, 28, 35, and a final timepoint (70 days for 2014 and 31–33 days in 2015) post-lesion creation using a Nikon D7100 with a Sigma 8–16-mm wide-angle lens in an Ikelite housing. Each lesion photographed included a tape measure immediately adjacent to the lesion to act as a scale bar. Images were used to quantify the number of polyps incidentally damaged during lesion creation and colony colour. Images were imported to ImageJ (Abramoff et al. 2004) where the scale bar to pixel ratio for each image was calibrated and the surface area and perimeter of each lesion was determined using the freehand line tool. The rate of change of lesion size between each sampling timepoint for each individual coral was estimated using the approximation formula:

$$-b = \frac{2(y_1 - y_2)}{(y_1 + y_2)}$$

and these estimated rates were tested using repeated measures ANOVA with island, site, colour of the colony, and number of polyps damaged, as potential explanatory

Table 1 Site names, locations, GPS coordinates, year and number of colonies that received lesions in the nearshore waters of Grenada and Carriacou, Grenada, West Indies

Island	Site name	Year	No. of colonies	Proteomics	GPS coordinates
Grenada	Dragon Bay	2014	16	N	12°5′6.00″N 61°45′45.36″W
Grenada	Flamingo Bay	2014	14	N	12°5′30.36″N 61°45′30.60″W
Grenada	Quarter Wreck	2014	14	N	12°1′40.98″N 61°47′0.84″W
Grenada	Northern Exposure	2014	16	N	12°2′22.14″N 61°46′4.74″W
Carriacou	Sandy Island	2014	16	N	12°29′7.77″N 61°29′2.00″W
Carriacou	Whirlpool	2014	16	N	12°29′2.50″N 61°29′31.35″W
Carriacou	Jackadam	2014	16	N	12°29′46.25″N 61°28′4.11″W
Carriacou	Seaview	2014	16	N	12°26′44.13″N 61°29′59.70″W
Carriacou	Sandy Island	2015	10	Y	12°29′7.77″N 61°29′2.00″W
Carriacou	Whirlpool	2015	10	N	12°29′2.50″N 61°29′31.35″W
Carriacou	Jackadam	2015	10	Y	12°29′46.25″N 61°28′4.11″W

factors and covariates. Nonlinear least squares regression was then used to fit an exponential model to the current size of the lesion with the factors that were significant in the repeated measure model as separate predictors in addition to the number of days post-lesion creation. Repeated measures ANOVA was used to determine whether there was a significant difference in the rate of change for lesions created on Carriacou between years. Once statistical models of lesion closure had been developed, they were used to estimate when rapid changes in lesion size were occurring. Based on these models, tissue sampling times were selected for proteomic analysis to capture changes at the proteomic level. Tandem mass tag (TMT)-labelled liquid chromatography–mass spectrometry was used to determine which proteins were DA in *M. cavernosa* during the first month (days 0–33) of the regenerative process. All statistical analyses were conducted using R Version 3.4 (R Core Team 2017) and ggplot (Wickham 2016) was used to create figures.

Sample collection for proteomic analysis

Coral polyp core samples were collected from lesions created haphazardly on the tops of targeted *M. cavernosa* colonies at two previously established study sites (Sandy Island and Jackadam) in the nearshore waters of Carriacou in 2015 (Table 1).

Four lesions were created on each colony (Fig. 1), each of which was separated from any other lesion and the bottom of the colony by at least 8 cm. One of the original four lesions was re-sampled at 14 days post-lesion creation, another at 21, and a third at a final timepoint (day 32 or 33) and processed and frozen as described above for proteomic analysis (Fig. 2). The final tissue collection timepoint varied from day 32 to 33 due to logistical reasons. The collection protocol allowed for tissue to be sampled initially and at two stages during regeneration and at the end of the first month of the regeneration period. Lesions were re-sampled by re-drilling the same core that was originally created and collecting the regenerated tissue healing inwards from the periphery of the lesion.

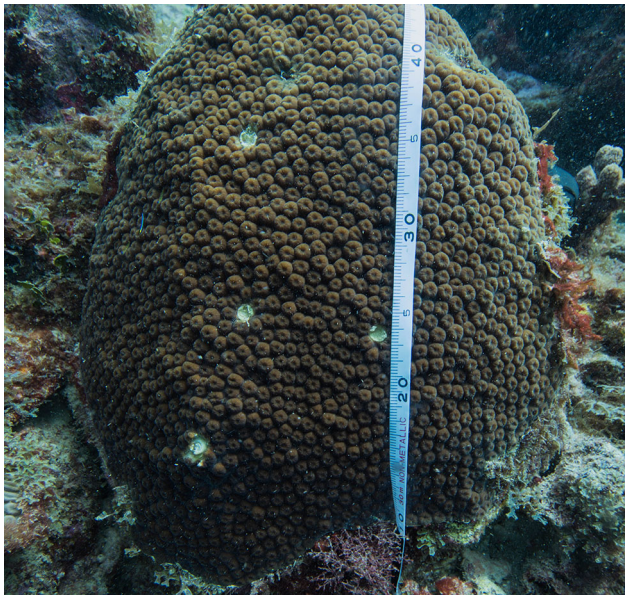

Fig. 1 *Montastrea cavernosa* colony from 2015 at Jackadam, Carriacou with four lesions created

Tissue samples were collected for proteomic analysis during regeneration from five brown, normally pigmented *M. cavernosa* colonies at days 0, 14, 33 at Sandy Island and from five brown, normally pigmented *M. cavernosa* colonies at days 0, 14, 32 at Jackadam. These colonies were selected for proteomic analysis based on the estimated lesion regeneration rates ($-b$) for individual lesions. Coral colonies selected for proteomic analysis had estimated lesion regeneration rates ($-b$) within one standard deviation of each other. Brown colonies were selected because they are the most commonly observed colour morph on the nearshore reefs of Carriacou.

Soluble protein extraction

Protein was extracted from coral polyp tissue that had been stored at − 80 °C following the protocol established in Barneah et al. (2006). Briefly, tissue was removed from the skeleton using a sterile scalpel blade and placed in a 1.5-ml microcentrifuge tube on ice with 250 µl of extraction buffer (40 mM Tris, 10 mM EDTA, pH 7.4) and 250 µl protease inhibitor cocktail (Sigma-Aldrich, Oakville, ON). Tissue was homogenized in the extraction buffer on ice for 30 s using a sterile plastic pestle and then inverted gently for 30 s to mix. The homogenate was centrifuged for 20 min at 12,000×*g* at 4 °C. The supernatant was transferred to a new microcentrifuge tube and centrifuged for 20 min at 12,000×*g* at 4 °C two additional times. Protein concentrations were determined using a modified Bradford assay in a 96-well plate (Bradford 1976).

Mass spectrometry

Extracted protein samples were submitted on dry ice to the SPARC Bio Centre at the Toronto Hospital for Sick Children for tandem mass tag labelling and mass spectrometry using the Q-Exactive protocol on a ThermoFisher Orbitrap liquid chromatography–mass spectrometry (LC–MS) device (ThermoFisher, San Jose, CA).

The proteins were reduced in 1 mM dithiothreitol (DTT) for 1 h at 56 °C and the free cysteine residues were alkylated using iodoacetamide. The proteins were precipitated with 5 volumes of acetone overnight at 20 °C and

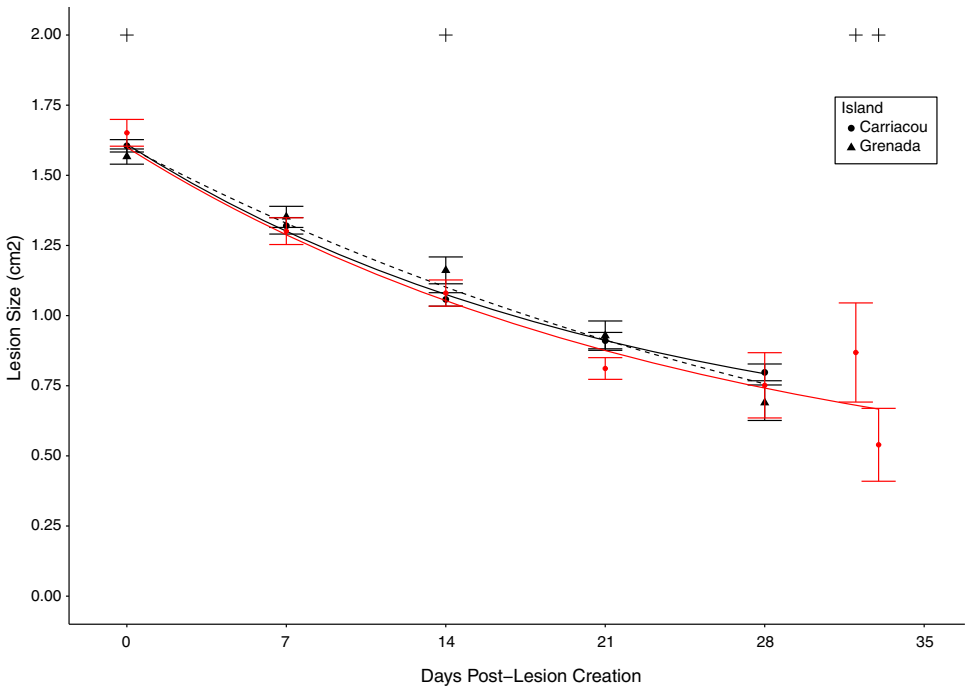

Fig. 2 Regression models (nonlinear least squares) for the reduction in *Montastraea cavernosa* lesion size (cm^2 with standard error) over a 35-day period from September to November 2014 for Grenada (dashed line), September to November 2014 for Carriacou (solid line), and November to December 2015 for Carriacou (solid red line). Sample collection dates are noted (+)

centrifugation. 10 µg of protein from each condition was labelled using 0.4 mg of TMT 10-plex by incubating at room temperature for 1 h. The labelling reaction was stopped using 5% hydroxylamine. The peptides were mixed, and the solvent removed under vacuum. The pellet was re-suspended, and samples were analysed on a Orbitrap analyser (Q-Exactive, ThermoFisher) outfitted with a nanospray source and EASY-nLC nano-LC system (ThermoFisher). Lyophilized peptide mixtures were dissolved in 0.1% formic acid and loaded onto a 75 µm × 50 cm PepMax RSLC EASY-Spray column (ThermoFisher) filled with 2 µM C18 beads (ThermoFisher) at a pressure of 800 bar. Peptides were eluted over 120 min at a rate of 250 nl min^{-1} using a stepwise gradient (0–4% acetonitrile containing 0.1% formic acid over 2 min; 4–28% acetonitrile containing 0.1% formic acid over 106 min, 28–95% acetonitrile containing 0.1% formic acid over 2 min, constant 95% acetonitrile containing 0.1% formic acid for 10 min). Peptides were introduced by nano-electrospray into the Q-Exactive mass spectrometer (ThermoFisher). The instrument method consisted of one MS full scan (525–1600 m/z) in the Orbitrap mass analyser with an automatic gain control (AGC) target of 1e6, maximum ion injection time of 120 ms and a resolution of 35 000 followed by 15 data-dependent MS/MS scans with a resolution of 35,000, an AGC target of 1e6, maximum ion time of 120 ms, and one microscan. The intensity threshold to trigger an MS/MS scan was set to an under-fill ratio of 0.2%. Fragmentation occurred in the higher energy collisional dissociation trap with normalized collision energy set to 30 V. The dynamic exclusion was applied using a setting of 40 s. Four separate LC–MS runs were used to acquire data for this study. Two separate runs were used for the samples originating from Sandy Island, and two separate runs were used for the samples originating from Jackadam. Day 0, 14, and 31 samples from an individual colony were always in the same run.

Data analysis

Raw files were analysed at the University of Guelph using PEAKS Studio 8.0 (Bioinformatics Solutions Inc., Waterloo, ON) with TMT 10-plex and carbamidomethylation set as fixed modifications, and oxidation and deamidation as variable modifications. A digestion enzyme of trypsin with three maximum missed cleavages was used. The data were searched with a parent ion tolerance of 35.0 ppm and a fragment error tolerance of 0.05 Dalton (Da). De novo assemblies were made in PEAKS, and database searches of the *M. cavernosa* transcriptome made available by Kitchen et al. (2015) were conducted. Peptide results were filtered for each run with a false discovery rate (FDR) of 1%. Protein results were filtered with a − 10 LgP score of ≥ 15 and at least 2 unique peptides. The de novo only average local confidence (ALC) was set for each run following the protocol described in the PEAKS user manual (PEAKS Team 2013), typically ≥ 60%.

Using a common TMT-labelled sample allowed for inter-run normalization. PEAKS quantification, based on the MaxQuant algorithm (Cox and Mann 2008), was used

to assess the change in protein abundance between sample timepoints. Spectra with a − 10 LgP score ≤ the 1% FDR and a quality score ≤ 16 were excluded from the analysis. Following the PEAKS user manual (PEAKS Team 2013), a PEAKS significance cut-off value of ≥ 15 using PEAKSQ (Cox and Mann 2008), a fold change of ≥ |1|, and at least two unique peptides were used to identify proteins that were DA within each run. Individual TMT run results were combined using the global normalization in PEAKS and in the 24 cases of DA proteins common to more than one TMT run, the intensity values for the reporter ions were averaged. The basic local alignment search tool (BLAST; Altschul et al. 1990) was used to search the anthozoan database of the National Centre for Biotechnology Information (NCBI) with an expectation value (e value) of $1e{-}03$. Back-translated sequences of all DA proteins were submitted to the NCBI BLAST service to confirm that they had originated from coral tissue. Protein accession IDs from the TMT results were exported as.fasta files to BLAST 2 GO (Conesa et al. 2005). BLAST hit gene identifiers were used to retrieve gene ontology (GO) terms and confidence scores (Conesa et al. 2005). The GO terms were added to the BLAST hits in BLAST 2 GO (Conesa et al. 2005). To determine major GO groupings, GO term categories needed to contain at least 10% of the total number of sequences in the dataset.

Results

Lesion regeneration

All lesions created in 2014 (n = 128) were 1.587 ± 0.193 cm^2 (average ± SE) at the time of creation. There were no significant differences for repeated measures ANOVA in the mean rates of change of lesion size between sites, colony colour, or number of incidental polyps damaged in 2014 ($p > 0.05$). There were significant differences for repeated measures ANOVA in the mean rates of change of lesion size between islands in 2014 (p = 0.024); therefore, separate lesion regeneration curves were generated for Grenada and Carriacou in 2014 (Table 2, Fig. 2).

All lesions created in 2015 (n = 124) were 1.598 ± 0.261 cm^2 (average ± SE) at the time of creation. There were no significant differences for repeated measures ANOVA in the mean rates of change of lesion size between sites or number of incidental polyps damaged in 2015 ($p > 0.05$); however, there were significant differences between colony colour (p = 0.024). Each lesion was surrounded by 4.52 polyps ± 1.15, and there were no significant differences in the number of polyps surrounding a lesion between islands (p = 0.566). Using repeated measures ANOVA, there were no significant differences in regeneration rate between lesions created on Carriacou in 2014 and 2015 ($p > 0.05$), and the confidence intervals for estimated parameters for the models generated for 2014 and 2015 overlapped. While the data for each individual year generated different models, the overlap of confidence intervals for the predictors suggests there are no significant differences between lesion regeneration models for each year on Carriacou (Table 2, Fig. 2). A separate model created for only the lesions used in the proteomic analysis was not significantly different than the model for all lesions on Carriacou in 2015 ($p > 0.05$, Fig. 2).

Table 2 Exponential decay models for *M. cavernosa* lesion regeneration in Carriacou and Grenada in 2014 and 2015 with significance, estimate of variable, and standard error (in parenthesis)

Year	Model		Variable	p Value
2014	GND: $y = a e^{-b*\text{time}}$	a	1.604 (0.041)	$< 2e{-}16$*
		$-b$	0.027 (0.002)	$< 2e{-}16$*
2014	CAR: $y = y_0 + a e^{-b*\text{time}}$	y_0	0.478 (0.150)	$1.60e{-}3$*
		a	1.133 (0.142)	$3.19e{-}14$*
		$-b$	0.046 (0.011)	$2.61e{-}05$*
2015	CAR: $y = a e^{-b*\text{time}}$	a	1.628 (0.047)	$< 2e{-}16$*
		$-b$	0.029 (0.002)	$< 2e{-}16$*

*Denotes significant differences from 0 at α = 0.05

Gross lesion regeneration

The wounds healed inwards from the damaged tissue on the periphery of the circular lesions. A layer of unpigmented tissue progressed inwards to cover the exposed skeleton until the entire lesion was infilled (Figs. 3, 4). After the lesions were completely resealed by unpigmented tissue, a polyp mouth became visible (Fig. 5). Finally, normal tissue pigmentation returned to the regenerated polyps. Eight per cent of all artificially induced lesions regenerated tissue and a mouth within the first 35 days in 2014 while 35% of all lesions regenerated tissue and a mouth within the total 70-day observation period in 2014. Twelve per cent of all lesions regenerated tissue and a mouth within the 35-day observation period in 2015. None of the regenerated lesions regained complete pigmentation during the observation period in 2014 or 2015. All lesions that were created for this study healed completely within 2 months after the observation periods in 2014 and 2015 had ended. Complete lesion regeneration was confirmed visually by personnel employed by dive operators. These same colonies were assessed visually for scar formation in 2016 and 2017, and no scarring was present.

Fig. 3 *Montastraea cavernosa* colony from Jackadam in Carriacou, Grenada on the day of lesion creation (day 0) with excision of a single polyp and peripheral damage to 5 other polyps

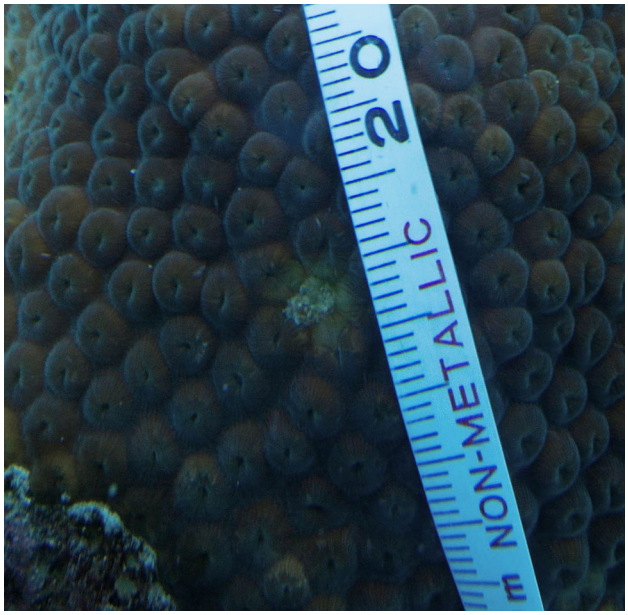

Fig. 4 *Montastraea cavernosa* colony from Jackadam in Carriacou, Grenada 14 days after initial lesion creation with depigmentation of the damaged peripheral polyps and proliferating tissue replacing the lost polyp

Detection of protein profiles during regeneration

Of the 6419 peptides reliably identified using the criteria established in the materials and methods section, 906 unique proteins were identified (Supplementary Tables). Using a sequence filter of 10% of sequences in Blast 2 GO, 54% of the proteins detected in this study, both DA and otherwise, were linked to either cellular or protein metabolic processes. Thirty-three per cent of identified proteins were associated with single-organism processes. The remaining 13% of identified proteins could be linked to biological regulation, cellular organization or biogenesis, response to stimulus, signalling, development, adhesion, and detoxification.

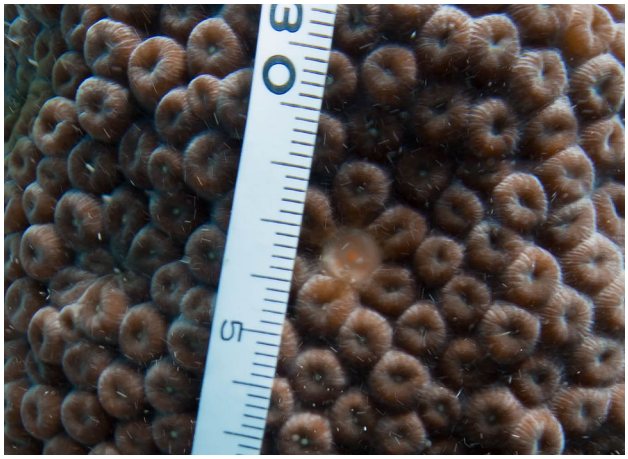

Fig. 5 *Montastraea cavernosa* colony at Sandy Island in Carriacou, Grenada, 31 days after initial lesion creation. Note the orange mouth at the centre of the healed polyp and slight pigmentation of the regenerated tissue

For presentation purposes the proteins that were DA in the regenerating tissue of wounded *M. cavernosa* are grouped into those associated with regeneration, inflammation, and antioxidant activity; however, many proteins play a role in multiple stages of regeneration. Since many of the protein identities were highly similar to better-characterized proteins (e.g. thioredoxin-like or L-rhamnose-binding lectin-like) in other phyla, the assignment of function is somewhat speculative. Most proteins that were DA were inflammatory, associated with tissue reorganization, or were classified as 'other'. The majority of DA proteins were also consistently altered at both timepoints during regeneration.

Nine hundred and six unique proteins were reliably identified in the present study. One hundred and eleven of these proteins were significantly ($p < 0.05$) DA during the regeneration process (Fig. 6). Twenty-four proteins were DA in at least two separate LC/MS runs (Table 3). Of these calmodulin, catalase, dipeptidyl-peptidase 5, ferritin, glutathione-requiring prostaglandin-D synthase and five unknown proteins were common to all TMT runs and all coral tissues used in the present study (Table 3). The back-translated nucleic acid sequences from all DA proteins were each submitted individually to the NCBI BLAST service. Of the five unknown proteins submitted to the NCBI BLAST service, four returned annotations for uncharacterized proteins, while one returned no known similar sequences (Table 3). In addition, 25 proteins

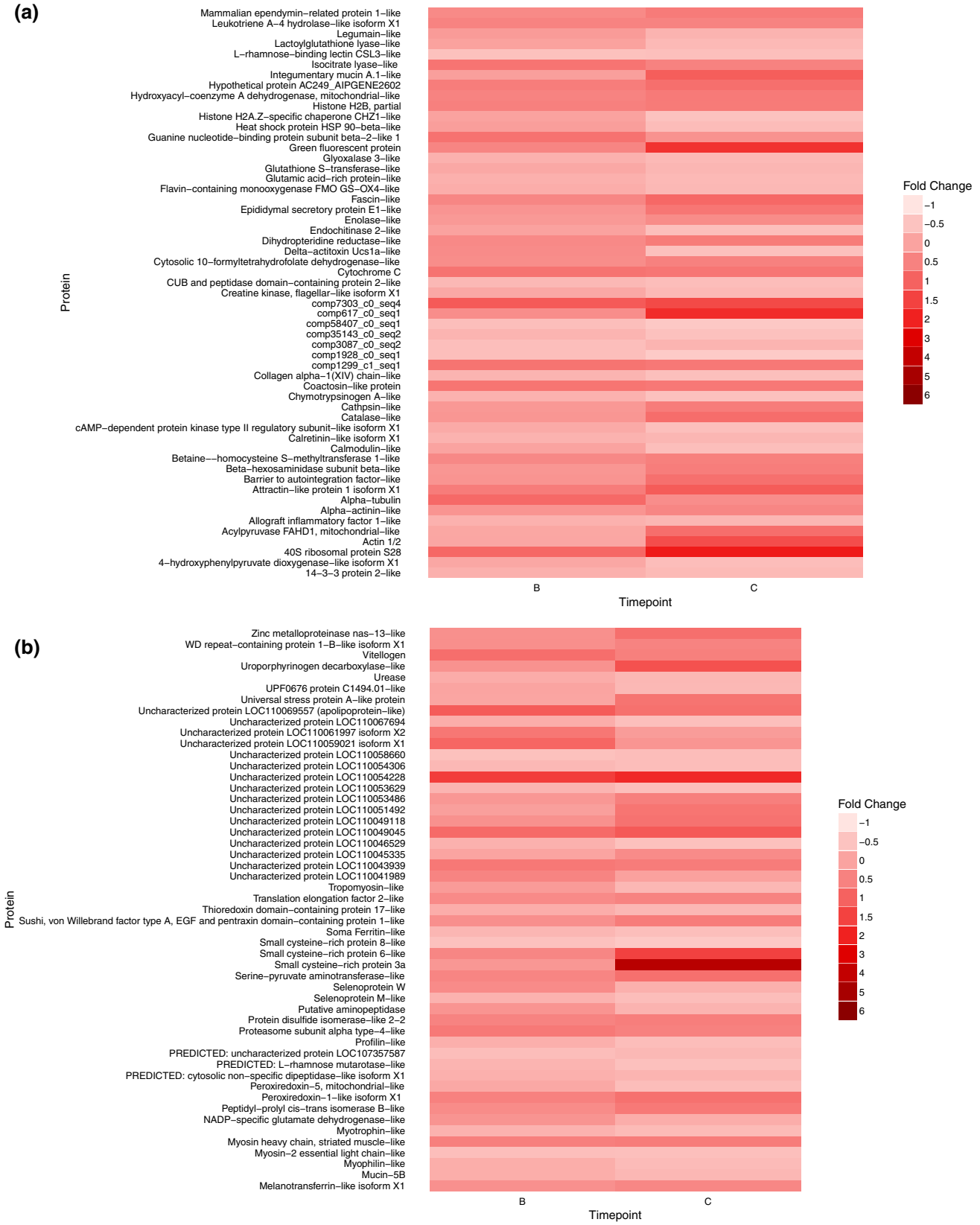

Fig. 6 Heatmap showing the fold change of all differentially abundant proteins at sampling timepoints day 14 (B) and day 32 or 33 (C) relative to day 0. Higher abundance is denoted by darker colours

Table 3 Proteins common to at least two LC–MS runs with fold change at day 14 (B) and day 32 or 33 (C) relative to day 0 that were differentially abundant during the first month of regeneration in *Montastraea cavernosa* collected in the nearshore waters of Carriacou, Grenada in 2015

UniProt	Protein	Gene	Organism	% Identity	*e* Value	Fold B	Fold C
J3NY69	Calmodulin-like	*GGTG_06222*	*O. faveolata*	96	6*e*−94	0.004	− 0.415
A0EJ86	Catalase-like	N/A	*O. faveolata*	93	0.0	0.203	0.820
R9NY46	Dipeptidyl-peptidase 5	*PHSY_001159*	*P. hubeiensis*	44	1*e*−63	0.583	0.797
K1RA71	Glutathione-requiring prostaglandin-D synthase	*CGI_10022749*	*O. faveolata*	66	3*e*−86	− 0.084	− 0.299
A7S8I6	Soma ferritin-like	*v1g235103*	*O. faveolata*	93	2*e*−103	− 0.287	− 0.422
K1R289	Collagen alpha-1(VI)-like	*CGI_10013230*	*O. faveolata*	76	4*e*−77	− 0.257	− 0.427
K1QB11	Myophilin-like	*CGI_10020262*	*O. faveolata*	85	1*e*−125	− 0.162	− 0.372
Q0ZDM0	Tropomyosin-like	N/A	*O. faveolata*	91	3*e*−145	0.117	− 0.314
A8WH03	Cytosolic 10-formyltetrahydrofolate dehydrogenase-like	N/A	*O. faveolata*	98	0.0	0.335	0.514
V5GJC6	Translation elongation factor 2-like	*EF2*	*A. digitifera*	87	0.0	0.381	0.488
E0XPT3	Flavin-containing monooxygenase FMO GS-OX4-like	N/A	*O. faveolata*	93	0.0	− 0.161	− 0.330
Q5XUI0	Peroxiredoxin-1-like isoform X1	N/A	*O. faveolata*	93	3*e*−167	0.522	0.769
N/A	Uncharacterized protein LOC110049118	N/A	*O. faveolata*	77	N/A	0.283	0.755
N/A	PREDICTED: uncharacterized protein LOC107357587	N/A	*A. digitifera*	53	N/A	− 0.400	− 0.327
T2MH98	Cathepsin-like	*CTSL1*	*O. faveolata*	91	0.0	0.162	0.595
Q3V5L7	Integumentary mucin A.1-like	*trf*	*A. digitifera*	43	6*e*−116	0.059	1.065
N/A	Uncharacterized protein LOC110046529	N/A	*O. faveolata*	86	N/A	− 0.211	− 0.454
N/A	LOC110066892	N/A	*O. faveolata*	66	N/A	0.410	0.636
H3II20	Peptidyl-prolyl cis–trans isomerase-B-like	N/A	*O. faveolata*	93	7*e*−140	0.353	0.647
D2XMR9	Mammalian ependymin-related protein 1-like	N/A	*O. faveolata*	89	2*e*−94	0.384	0.605
Q1XGE0	Guanine nucleotide-binding protein subunit beta-2-like-1	*GNB2L1*	*A. digitifera*	88	0.0	0.751	0.266
N/A	Uncharacterized protein LOC110041989	N/A	*O. faveolata*	65	0.0	0.477	0.037
N/A	comp617_c0	N/A	N/A	N/A	0.0	0.332	1.837
C1KIZ1	Small cysteine-rich protein 3a	N/A	*O. faveolata*	53	6*e*−24	0.187	4.378

O Orbicella, *P Pseudozyma*, *A Acropora*

e Values smaller than 2.225074*e*−308 are represented as 0.0

associated with regeneration, inflammation, or immunity were also DA during regeneration (Table 4).

Discussion

The first objective of the present study was to test the hypothesis that lesion size of *M. cavernosa* colonies would decrease exponentially to an asymptote determined by the initial lesion size as observed in other coral species (proposed by Fisher et al. 2007; Meesters et al. 1997). The lesions created on *M. cavernosa* colonies in the present study decreased exponentially in size in a manner similar to that observed in other coral species, although only reached an asymptote in Carriacou in 2014. Once regeneration rates, and therefore appropriate intervals for tissue sampling, had been determined, the proteomic mechanisms of *M. cavernosa* tissue regeneration were examined. This was accomplished through observing the differential abundance of protein during *M. cavernosa* tissue regeneration using TMT-labelled samples and LC–MS.

Lesion regeneration

Two separate models describing lesion closure with time were generated using the digital images collected in the first month of regeneration in 2014 and 2015. These models were highly similar which suggests that the data collected

Table 4 Fold change at Day 14 (B) and Day 32 or 33 (C) relative to Day 0 and identity of proteins associated with regeneration or inflammation during the first month of regeneration in *Montastraea cavernosa* collected in the nearshore waters of Carriacou, Grenada in 2015

UniProt ID	Protein	Function	Gene	Organism	% Identity	*e* Value	Fold B	Fold C
O61375	Actin 1 (Fragment)	Reg.	*DpAct1*	*E. pallida*	100	3*e*−46	− 0.04	1.32
Q0H8V2	Allograft inflammatory factor	Inf.	N/A	*O. faveolata*	96	1*e*−79	− 0.21	− 0.30
J3NY69	Calmodulin	Inf.	*GGTG_06222*	*O. faveolata*	96	6*e*−94	0.00	− 0.42
A0EJ86	Catalase	Inf.	N/A	*O. faveolata*	93	0.0	0.20	0.82
E0XJK4	Heat shock protein 90	Reg./Inf.	N/A	*O. faveolata*	98	0.0	0.08	− 0.39
Q9HC84	Mucin-5B	Inf.	*MUC5B*	*E. pallida*	36	0.0	− 0.16	− 0.30
T2MHB9	Alpha-enolase	Inf.	*ENO1*	*E. pallida*	83	0.0	0.15	0.36
B2ZG38	Small cysteine-rich protein 8	Inf.	N/A	*O. faveolata*	67	4*e*−24	− 0.47	− 0.57
C1KIY9	Small cysteine-rich protein 6	Inf.	N/A	*O. faveolata*	39	0.001	0.44	1.51
C1KIZ1	Small cysteine-rich protein 3a	Inf.	*SCRiP3a*	*O. faveolata*	53	6*e*−24	0.19	4.38
K1QJC6	Coactosin-like protein	Reg.	*CGI_10027046*	*A. digitifera*	75	3*e*−65	0.71	0.66
J7QAM2	Cytochrome C	Reg.	*cytC*	*O. faveolata*	95	9*e*−68	0.74	0.69
V5GJC6	Elongation factor 2	Reg.	*EF2*	*A. digitifera*	97	0.0	0.38	0.49
O96899	Plasminogen activator sPA	Reg.	N/A	*O. faveolata*	79	2*e*−141	− 0.32	− 0.41
P18321	Profilin	Reg.	N/A	*E. pallida*	80	4*e*−74	− 0.18	− 0.40
L7M2A4	Putative alpha actinin	Reg.	N/A	*O. faveolata*	97	0.0	0.23	0.45
Q5XUI0	Thioredoxin peroxidase	Reg.	N/A	*B. belcheri*	77	1*e*−104	0.52	0.77
Q6PAX7	WD repeat-containing protein 1-B	Reg.	*wdr1-b*	*O. faveolata*	86	0.0	0.31	0.49
T2MH98	Cathepsin L1	Reg.	*CTSL1*	*O. faveolata*	94	0.0	0.16	0.60
Q962P9	Green fluorescent protein	Ant.	N/A	*O. faveolata*	87	7*e*−137	0.46	1.75
T2ME17	Zinc metalloproteinase nas-13-like	Ant.	*ALKBH1*	*O. faveolata*	84	6*e*−46	− 0.04	0.29
Q5XUI0	Peroxiredoxin-1-like isoform X1	Ant.	N/A	*O. faveolata*	93	3*e*−167	0.52	0.77
L5KNS3	Peroxiredoxin-5-like	Ant.	*GLEAN10011419*	*O. faveolata*	87	5*e*−22	− 0.07	− 0.40
S0GP72	L-rhamnose-binding lectin CLS3-like	Ant.	*C803_04027*	*O. faveolata*	88	2*e*−65	− 0.26	− 0.45
T2MHF5	Apolipoprotein-like	Ant.	*APOB*	*O. faveolata*	93	0.0	1.11	0.77

e Values smaller than 2.225074*e*−308 are represented as 0.0

Inf. Inflammation; *Reg.* Regeneration; *Ant.* Antioxidant; *E Exaipasia*; *O Orbicella*; *A Acropora*; *B Branchiostoma*

on lesion regeneration in the first month are accurate predictors of lesion closure in coral colonies surveyed at similar depths (Fisher et al. 2007). The differences in rate of regeneration between Grenada and Carriacou in 2014 may have been the result of individual colony variation although great care was taken to sample colonies of approximately the same size. Colony reproductive status, colony fecundity, and lesion regeneration have been linked in a number of studies (Graham and van Woesik 2013; Kramarsky-Winter and Loya 2000; Van Veghel and Bak 1994). Lesions created in Carriacou in October of 2015 produced regeneration curves that were not significantly different ($p > 0.05$) from those created in Grenada in September of 2014, despite *M. cavernosa* spawning in September.

The rates of regeneration documented in the present study for *M. cavernosa* (2.8 mm^2 d^{-1}) are slightly lower than those previously published for the *O. annularis* species complex (Table 5). Most regeneration studies in scleractinia that have used a standardized lesion have created lesions that are irregularly shaped or very long and narrow, whereas those created in the present study were cylindrical. Importantly, it has been demonstrated that circular lesions heal more slowly than irregularly shaped lesions (Meesters et al. 1997). Most studies report different mean regeneration rates between species that vary with the overall size of the colony (Bak and Meesters 1998; Henry and Hart 2005; Kramarsky-Winter and Loya 2000), shape of the lesion (Meesters et al. 1997) and overall colony morphology (Bak and Meesters 1998; Hall 1997; Kramarsky-Winter and Loya 2000). Since *M. cavernosa* is a massive/submassive species, it was reasonable to have observed a slower lesion regeneration rate for this type of colony morphology, size of colony, and shape of lesion. Finally, Meesters et al. theorized that there is a finite amount of tissue that can be regenerated and the amount is

Table 5 Estimated daily regeneration rates of *Montastraea annularis and Orbicella annularis* (adapted and updated from Henry and Hart 2005)

Species	Lesion size (mm^2)	Daily regeneration rate (mm^2)	References
Montastraea annularis	22.5–502	10–90	Croquer et al. (2002)
Montastraea annularis	227	2.9–3.6	Mascarelli and Bunkley-Williams (1999)
Montastraea annularis	100	1.9	Meesters et al. (1992)
Montastraea annularis	100–500	0.75–0.94	Bak et al. (1977)
Montastraea annularis	7.5–30.2	0.0013	Fisher et al. (2007)
Montastraea annularis	170	4.1	Meesters et al. (1994)
Montastraea annularis (bumpy morph)	160	15.4–18.7	Van Veghel and Bak (1994)
Montastraea annularis (columnar morph)	83–406	3.7–11.2	Meesters et al. (1997)
Montastraea annularis (columnar morph)	160	0.7	Van Veghel and Bak (1994)
Montastraea annularis (massive morph)	79	1.6	Meesters and Bak (1993)
Montastraea annularis (massive morph)	160	10.5–13.9	Van Veghel and Bak (1994)
Orbicella annularis	31.4	Not given	Rodríguez-Martínez et al. (2016)
Orbicella annularis	37 ± 0.9	11.1 ± 0.88	Sabine et al. (2015)

governed by the ratio of the lesion perimeter to its surface area (Meesters et al. 1994, 1997). Contrary to this, no significant differences were found in the estimate of regeneration rate ($-b$) that could be attributed to the number of polyps surrounding a lesion (data not shown). This is likely the result of the low lesion complexity and low variability (± 0.193 cm^2 SE; 12%) with the standardized circular lesions that were created in the present study.

Differential protein abundance

Many proteins that are involved in the wound-healing process of both vertebrates and invertebrates were DA during the regeneration of tissue and skeletal lesions created on *M. cavernosa* colonies. This suggests that much like the healing of wounds at the cellular level (Palmer et al. 2011), scleractinian corals use similar proteomic mechanisms as vertebrates to completely regenerate damaged tissue. The proteins that were DA in regenerating *M. cavernosa* tissue have been separated into the following categories for ease of discussion; coagulation, inflammation, proliferation, maturation, reorganization, immune-related, and others.

Coagulation

No proteins that could be reliably linked to coagulation were detected in the regenerating tissues of *M. cavernosa*. This is likely the result of the times that tissue samples were collected for proteomic analysis in the present study. Palmer et al. (2011) noted increased concentrations of amoebocytes at the periphery of lesions just 6 h after their creation. The lesions created on *M. cavernosa* colonies had insufficient tissue for sampling at a timepoint earlier than 14 days and what was present would have been inadequate for proteome analysis. Samples at earlier timepoints pooled from multiple corals would be required to obtain DA proteins using the present methodology. Logistically, this was not possible in the present study; however, this could be performed in the future.

Inflammation

There were numerous DA proteins that have a demonstrated or putative role in the modulation of inflammation. Allograft inflammatory factor-1 is a member of a family of proteins in activated macrophages in mammals and that have a role in calcium binding (Deininger et al. 2002). This protein was detected in reduced abundance in regenerating *M. cavernosa* tissue (fold change (FC) − 0.21/− 0.30). Calmodulin also binds calcium and in corals is highly structurally similar to calmodulins from most other phyla and also has the same exon/intron arrangement (Chiou et al. 2008). After binding calcium, calmodulin then also binds other proteins modulating numerous functions including protein kinases (reviewed by Swulius and Waxham 2008). It therefore acts as an intermediary protein in the signalling pathways for many important biological processes such as inflammation (Zhang et al. 2011), metabolism, apoptosis, smooth muscle contraction, intracellular movement, and the immune response (Stevens 1982). Calmodulin in *M. cavernosa* was detected in reduced concentration as regeneration progressed (FC 0.004/− 0.415).

Catalase activity was documented in *M. cavernosa* tissue (Mydlarz and Palmer 2011), and a catalase-like protein had increased abundance during regeneration in the present study (FC 0.20/0.82). Proteins exhibiting presumed similar functions in *M. cavernosa* (peroxiredoxin-1-like isoform X1 and thioredoxin domain-containing protein 17-like, respectively) were all consistently detected at increased abundance during regeneration of *M. cavernosa* reflecting the increased need for control of free radicals in inflamed and regenerating tissue (Mydlarz and Palmer 2011). Reactive oxygen species (ROS) can upset biological processes through direct interaction with proteins as well as through the degradation of DNA (reviewed by Dalton et al. 1999; Lesser 2006). To combat the toxic effects of ROS, corals and other organisms employ numerous strategies such as the production of ferritin and zinc metalloproteinase (Császár et al. 2009). Soma ferritin-like protein, part of the large ferritin-like superfamily, binds iron or modulates iron transport (Andrews 2010). Gene expression was altered in larval corals in response to thermal stress (Polato et al. 2010), and the protein was detected at reduced abundance at both timepoints in the present study (FC − 0.287/− 0.422). This seems counterintuitive but might reflect exhaustion of this protein in inflamed tissues. Additionally, catalase, peroxiredoxin, and thioredoxin peroxidase are critical as antioxidants in inflamed or stressed coral tissue (Chelikani et al. 2004; Fuess et al. 2016; Lesser and Shick 1989; Mydlarz and Palmer 2011; Sunagawa et al. 2009a, b; Van Oppen and Gates 2006).

Heat shock proteins (HSP) are molecular chaperones that maintain protein folding and are commonly altered via induction, in thermal and other coral stress responses (Barshis et al. 2010; Chen et al. 2006; Van Oppen and Gates 2006), and HSP90 was significantly increased and then decreased in the present study (FC 0.08/− 0.39). HSPs have been well studied in scleractinian corals, typically in response to elevated seawater temperatures from artificial and naturally bleached tissues (Brown et al. 2002; Chow et al. 2012; Downs et al. 2000; Fitt et al. 2009; Mayfield et al. 2011; Robbart et al. 2004) but also in immunity as gene expression of HSP 90 was induced in *O. faveolata* exposed to lipopolysaccharides from *E. coli* (Fuess et al. 2016). HSPs in the corallimorpharian *Corynactis californica* and the aggregating anemone (*Anthopleura elegantissima*) play a role in maintaining proper protein folding in response to physical stress (Rossi and Snyder 2001). HSPs were also increased in the presence of xenobiotics in *M. annularis* (Downs et al. 2005) and disease in *A. muricata* (Seveso et al. 2012).

Proliferation

There was only one DA protein in the present study that was likely involved with cell proliferation; guanine nucleotide-binding protein subunit beta 2-like-1, also known as RACK-1. RACK-1 is a receptor of activated C kinase 1 that binds and stabilizes protein kinase C and influences cell growth (Chang et al. 2001). In *M. cavernosa*, this protein was most abundant after 2 weeks of tissue regeneration when the rate of lesion closure, and therefore potentially the rate of cell proliferation, was greatest (FC 0.751/0.266). Approximately 15% of proteins detected in the present study could be linked to regenerative processes, while nearly 20% were linked to either regeneration, growth, or the regulation of these processes and yet only a single protein associated with cell proliferation was DA. Given this discrepancy, it is likely that some of the proteins that influence proliferation were masked in the quantitative analysis or were DA to such a minor degree that they were not statistically significant. One possible explanation for this discrepancy would be the method by which tandem mass tags function. TMT quantifies the relative change in protein abundance between samples (Thompson et al. 2003). If certain proteins, such as cytokines, are expressed in very small quantities, then it is likely that shifts in the abundance of these proteins would be masked. Additionally, cytokines are expressed at very low (pg/ml) concentrations and are difficult to detect using mass spectrometry approaches (Stenken and Poschenrieder 2015). ELISA-based assays could be used to detect the presence of anthozoan cytokines but would need to be developed first.

Maturation and reorganization

The majority of DA proteins detected in this category are part of the extracellular matrix (ECM) (e.g. collagen a-1) or cytoskeleton (actin 1) and are involved in remodelling fibrin or extracellular matrix (plasminogen activator protein, peptidyl-prolyl cis–trans isomerase-B-like, zinc metalloprotease) or modifying the cytoskeleton (profiling, a-actinin, tropomyosin, myophilin, coactosin-like protein, WD repeat-containing protein 1B). Actins, tropomyosins, and tubulins are all involved in the diurnal regulation of coral tissue (Mayfield et al. 2010) and are predicted to be involved in the reorganization of coral tissue during regeneration. Cyclic AMP kinase and creatine kinase both decreased in abundance over time, and while this is somewhat surprising, not all kinases act the same in coral as they do in mammals (Voolstra et al. 2011). Elongation factor 2 (EF2) is involved in polypeptide elongation, required for protein synthesis, and is highly conserved in eukaryotes (Baldauf et al. 1996). The increased abundance

of EF2 at both timepoints in *M. cavernosa* undergoing regeneration (FC 0.38/0.49) is consistent with the demand for increasing protein synthesis.

The ECM proteins collagen, elastin, fibronectin, and laminin are used for cell to cell connections (Alberts et al. 2002) and were all DA in the present study. The ECM of *Montipora digitata* in cell culture stained positively for collagen (Helman et al. 2008), and it has been identified in the skeletal organic matrix (SOM) of *Stylophora pistillata* (Drake et al. 2013). The SOM is considered integral in the deposition of the aragonite skeleton of scleractinian corals (Goffredo et al. 2011). Collagen (Goldberg 1974), cysteine-rich proteins (Sunagawa et al. 2009a, b), and von Willebrand factor type A domain-containing proteins and zona pellucidas (Hayward et al. 2011) are all hypothesized to play a role in skeleton deposition. These proteins were all DA in the present study consistent with skeletal deposition. Cathepsins are a group of proteases found in all animals that have also been linked to bone resorption in mammals (Teitelbaum 2000). A cathepsin-like protein was increased in abundance towards the end of the regenerative process in *M. cavernosa* (FC 0.16/0.60) also suggesting that skeletal tissue was being remodelled. Cathepsin L1 is a lysosomal endopeptidase associated with lysosomal protein turnover and apoptosis (Turk et al. 2012).

To promote lesion regeneration rather than scar formation, fibrotic tissues need to be degraded. Plasminogen degrades fibrotic tissue (Bryer et al. 2008; Novak et al. 2011), and a plasminogen activator decreased in abundance as the wound healed (FC − 0.32/− 0.41). Both cytochrome C (FC 0.74/0.69) and thioredoxin (FC 0.52/0.77) are linked to mediation of apoptosis and were DA as the tissue reorganization likely progressed.

Immune-related and others

Since the *M. cavernosa* wounds were exposed to the environment and would be considered contaminated, it is not surprising that putative immune proteins were also DA. Those of particular interest were two different mucins, green fluorescent protein, three small cysteine-rich proteins, an apolipoprotein-like protein, and L-rhamnose-binding lectin-like protein. Field observations have noted the copious production of mucus when corals are wounded, which may act as a primary barrier to infection (reviewed by Brown and Bythell 2005; Kramarsky-Winter and Loya 2000; Ritchie 2006; Shnit-Orland and Kushmaro 2009). The composition of coral mucus varies both qualitatively and quantitatively with species (Meikle et al. 1988), and the specific mucins within coral mucus may provide distinct services to the colony (Jatkar et al. 2010). Despite the critical importance of mucus to coral colony survival, very little is known about the specifics of its composition (Jatkar 2008). Cnidarian mucin gene sequences cluster together more closely within Cnidaria than with other species (Jatkar 2008), suggesting that there is strong divergence in cnidarian mucin genes, and potentially divergent function, from other organisms. Unique functions of coral mucins may help to explain the contrasting alterations of the two mucins (mucin 5B and integumentary mucin) detected in the present study (FC − 0.16/− 0.30, 0.059/1.065, respectively).

A green fluorescent protein had increased DA and was prominently more abundant at the final timepoint of regeneration in the present study (FC 0.46/1.75). A variety of fluorescent proteins from *O. annularis, O. faveolata, M. cavernosa, Diploria strigosa, Porites astreoides, Dichocoenia stokseii*, and *Sidastrea siderea* have been shown to act as antioxidants (Palmer et al. 2009). Green fluorescent proteins are also postulated to have a role in photoacclimation (Roth et al. 2010) and coral immunity (D'Angelo et al. 2012).

Small cysteine-rich proteins (SCRiPs) sequences have been noted in numerous scleractinian corals (Sunagawa et al. 2009a, b), and it is now recognized that one of their functions is as antimicrobial peptides (Closek et al. 2014). Of the three SCRiPs noted in the present study, two (SCRiP 6 and 3a) were increased (FC 0.44/1.51, 0.187/4.38) and one (SCRiP 8) was decreased in abundance (FC − 0.47/− 0.57). The increased abundance of SCRiP 6 and 3a over time in the present study may be an indication of a mounting immune response. None of the lesions observed in this study became fouled with visible bacterial communities as they regenerated, circumstantially supporting the evidence for the role of SCRiPs as antimicrobial peptides (Sunagawa et al. 2009a, b). SCRiP 3 gene activity has also been documented in the development of coral larvae (Siboni et al. 2014), and while larval development and tissue regeneration are distinct pathways, some of the required processes are the same (Burton and Finnerty 2009).

Both apolipoprotein (Di Natale 2017; Whitten et al. 2004) and L-rhamnose-binding protein (FC − 0.26/− 0.45) (Watanabe et al. 2009) are pattern recognition receptors (PRR) in fish and invertebrates binding lipid and rhamnose, respectively. Rhamnose-binding lectin in coral is a PRR that also binds pathogenic bacteria and zooxanthellae in *Pocillopora damicornis* (Zhou et al. 2017). In addition, two proteins associated with detoxification were also DA: alpha-enolase-like protein was upregulated (FC 0.15/0.36,) and glutathione-S transferase-like protein (Rougée et al. 2014) was downregulated (FC − 0.084/− 0.299). Finally, a mammalian ependymin-related protein 1-like protein, which was demonstrated to play a role in symbiosis of the anemone *Anemonia viridis* (Sabourault et al. 2009), was upregulated at both

timepoints in the present study (FC 0.384/0.605); however, attribution of any function during regeneration is speculative.

General

There was some variation in amount of tissue that was collected from each regenerating lesion, which could have influenced the abundance of certain proteins. The normalization process built into the analysis and the conservative cut-off values used would substantially limit the impact of any variation, however. This issue could be resolved in future experiments with precise sampling of a clonal laboratory population. The present study, however, was focused on examining the regeneration of lesions in the field in *M. cavernosa* as this captured the natural abundance of proteins.

Further analysis of the different colour morphs of *M. cavernosa* is planned to resolve the currently unaddressed differences in tissue regeneration rates observed between colonies of different colour and to provide additional data for whole proteome analysis. With the advent of marine proteomics and genomics, there are still many areas that require further investigation (Miller et al. 2011; Slattery et al. 2012). This study documents, for the first time, the differential abundance of proteins associated with regeneration, in a scleractinian coral. The results presented here confirm that the processes of wound healing and complete tissue regeneration are largely conserved across several distant metazoan species.

Acknowledgements We would like to thank Annette Patrick, Deefer Diving Carriacou, and EcoDive Grenada for their help with the collection of samples in the field, Bioinformatics Solutions Inc. and Dyanne Brewer for their technical assistance interpreting LC–MS data. Funding for this research was provided by a NSERC Discovery Grant (Lumsden) and St. George's University. Horricks received an OVC Scholarship.

Compliance with ethical standards

Conflict of interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

References

Abramoff MD, Magalhaes PJ, Ram SJ (2004) Image processing with imageJ. Biophotonics Int 11:36–41

Alberts B (ed) (2002) Molecular biology of the cell, 4th edn. Garland Science, New York, NY

Altschul SF, Gish W, Miller W, Myers EW, Lipman DJ (1990) Basic local alignment search tool. J Mol Biol 215:403–410

Anderson R, Morrall C, Nimrod S, Balza R, Berg C, Jossart J (2012) Benthic and fish population monitoring associated with a marine protected area in the nearshore waters of Grenada, Eastern Caribbean. Int J Trop Biol 60:71–87

Anderson R, Morrall C, Nimrod S, Balza R, Berg C, Jossart J (2014) Marine Protected Area monitoring in the nearshore waters of Grenada, Eastern Caribbean: benthic cover and fish populations. Rev Biol Trop 62:71–87

Andrews SC (2010) The Ferritin-like superfamily: Evolution of the biological iron storeman from a rubrerythrin-like ancestor. Biochim Biophys Acta 1800:691–705

Aurora AB, Olson EN (2015) Immune modulation of stem cells and regeneration. Cell Stem Cell 15:14–25

Bak RPM, Meesters EH (1998) Coral population structure: The hidden information of colony size-frequency distributions. Mar Ecol Prog Ser 162:301–306

Bak RPM, Steward-Van Es Y (1980) Regeneration of superficial damage in the Scleractinian corals *Agaricia agaricites* F. *Purpurea* and *Porites astreoides*. Bull Mar Sci 30:883–887

Bak RPM, Brouns JJWM, Heys FML (1977) Regeneration and aspects of spatial competition in the scleractinian corals Agaricia agaricites and Montastrea annularis. In: Proceedings of 3rd international coral reef Symposium Miami, pp 143–148

Baldauf SL, Palmer JD, Doolittle WF (1996) The root of the universal tree and the origin of eukaryotes based on elongation factor phylogeny. Proc Natl Acad Sci USA 93:7749–7754

Barneah O, Benayahu Y, Weis VM (2006) Comparative proteomics of symbiotic and aposymbiotic juvenile soft corals. Mar Biotechnol 8:11–16

Barshis DJ, Stillman JH, Gates RD, Toonen RJ, Smith LW, Birkeland C (2010) Protein expression and genetic structure of the coral *Porites lobata* in an environmentally extreme Samoan back reef: Does host genotype limit phenotypic plasticity? Mol Ecol 19:1705–1720

Bradford MM (1976) A rapid and sensitive method for the quantitation of microgram quantities of protein utilizing the principle of protein-dye binding. Anal Biochem 72:248–254

Brown BE, Bythell JC (2005) Perspectives on mucus secretion in reef corals. Mar Ecol Prog Ser 296:291–309

Brown BE, Downs CA, Dunne RP, Gibb SW (2002) Exploring the basis of thermotolerance in the reef coral *Goniastrea aspera*. Mar Ecol Prog Ser 242:119–129

Bryer SC, Fantuzzi G, Van Rooijen N, Koh TJ (2008) Urokinase-type plasminogen activator plays essential roles in macrophage chemotaxis and skeletal muscle regeneration. J Immunol 180:1179–1188

Burton PM, Finnerty JR (2009) Conserved and novel gene expression between regeneration and asexual fission in *Nematostella vectensis*. Dev Genes Evol 219:79–87

Chang BY, Chiang M, Cartwright CA (2001) The interaction of Src and RACK1 is enhanced by activation of protein kinase C and tyrosine phosphorylation of RACK1. J Biol Chem 276:20346–20356

Chelikani P, Fita I, Loewen PC (2004) Diversity of structures and properties among catalases. Arch Biochem Biophys 401:145–154

Chen B, Zhong D, Monteiro A (2006) Comparative genomics and evolution of the HSP90 family of genes across all kingdoms of organisms. BMC Genomics 7:156

Chiou CY, Chen IP, Chen C, Wu HJL, Wei NV, Wallace CC, Chen CA (2008) Analysis of *Acropora muricata* calmodulin (CaM) indicates that scleractinian corals possess the ancestral exon/intron organization of the eumetazoan CaM gene. J Mol Evol 66:317–324

Chow AM, Beraud E, Tang DWF, Ferrier-Pagès C, Brown IR (2012) Hsp60 protein pattern in coral is altered by environmental changes in light and temperature. Comp Biochem Physiol 161:349–353

Closek CJ, Sunagawa S, DeSalvo MK, Piceno YM, DeSantis TZ, Brodie EL, Weber MX, Voolstra CR, Andersen GL, Medina M (2014) Coral transcriptome and bacterial community profiles reveal distinct Yellow Band Disease states in *Orbicella faveolata*. ISME J 8:2411–2422

Conesa A, Götz S, García-Gómez JM, Terol J, Talón M, Robles M (2005) Blast2GO: A universal tool for annotation, visualization and analysis in functional genomics research. Bioinformatics 21:3674–3676

Cox J, Mann M (2008) MaxQuant enables high peptide identification rates, individualized p.p.b.-range mass accuracies and protome-wide protein quantification. Nat Biotechnol 26:1367–1372

Croquer A, Villamizar E, Noriega N (2002) Environmental factors affecting tissue regeneration of the reef-building coral *Montastraea annularis* (Faviidae) at Los Roques National Park, Venezuela. Revi Biol Trop 3:1055–1065

Császár NBM, Seneca FO, Van Oppen MJH (2009) Variation in antioxidant gene expression in the scleractinian coral *Acropora millepora* under laboratory thermal stress. Mar Ecol Prog Ser 392:93–102

D'Angelo C, Smith EG, Oswald F, Burt J, Tchernov D, Wiedenmann J (2012) Locally accelerated growth is part of the innate immune response and repair mechanisms in reef-building corals as detected by green fluorescent protein (GFP)-like pigments. Coral Reefs 31:1045–1056

Dalton TP, Shertzer HG, Puga A (1999) Regulation of gene expression by reactive oxygen. Annu Rev Pharmacol Toxicol 39:67–101

Deininger MH, Meyermann R, Schluesener HJ (2002) The allograft inflammatory factor-1 family of proteins. FEBS Lett 514:115–121

Di Natale A (2017) Apolipoprotein domain containing proteins and innate immunity in rainbow trout and walleye. MSc thesis, University of Guelph

Downs CA, Mueller E, Phillips S, Fauth JE, Woodley CM (2000) A molecular biomarker system for assessing the health of coral (*Montastraea faveolata*) during heat stress. Mar Biotechnol 2:533–544

Downs CA, Fauth JE, Robinson CE, Curry R, Lanzendorf B, Halas JC, Halas J, Woodley CM (2005) Cellular diagnostics and coral health: Declining coral health in the Florida Keys. 51:558–569

Drake J, Mass T, Haramaty L, Zelzion E, Bjattacharya D, Falkowski P (2013) *Proteomic* analysis of skeletal organic matrix from the stony coral *Stylophora pistillata*. Proc Natl Acad Sci 110:3788–3793

Fisher EM, Fauth JE, Hallock P, Woodley CM (2007) Lesion regeneration rates in reef-building corals *Montastraea* spp. as indicators of colony condition. Mar Ecol Prog Ser 339:61–71

Fitt WK, Gates RD, Hoegh-Guldberg O, Bythell JC, Jatkar A, Grottoli AG, Gomez M, Fisher P, Lajuenesse TC, Pantos O, Iglesias-Prieto R, Franklin DJ, Rodrigues LJ, Torregiani JM, van Woesik R, Lesser MP (2009) Response of two species of Indo-Pacific corals, *Porites cylindrica* and *Stylophora pistillata*, to short-term thermal stress: The host does matter in determining the tolerance of corals to bleaching. J Exp Mar Bio Ecol 373:102–110

Foster NL, Baums IB, Mumby PJ (2007) Sexual vs. asexual reproduction in an ecosystem engineer: The massive coral *Montastraea annularis*. J Anim Ecol 76:384–391

Fuess LE, Pinzón JH, Weil E, Mydlarz LD (2016) Associations between transcriptional changes and protein phenotypes provide insights into immune regulation in corals. Dev Comp Immunol 62:17–28

Galloway SB, Work TM, Bochsler VS, Harley RA, Kramarsky-Winters E, McLaughlin SM, Meteyer CU, Morado JF, Nicholson JH, Parnell PG, Peters EC, Reynolds TL, Rotstein DS, Sileo L, and Woodley CM (2007) Coral disease and health workshop: Coral histopathology II. NOAA Technical Memorandum NOS NCCOS 56 and NOAA Technical Memorandum CRCP 4. National Oceanic and Atmospheric Administration, Silver Spring, MD. 84p

Goffredo S, Vergni P, Reggi M, Caroselli E, Sparla F, Levy O, Goffredo S, Vergni P, Reggi M, Caroselli E, Sparla F, Levy O, Falini G (2011) The skeletal organic matrix from Mediterranean coral *Balanophyllia europaea* influences calcium carbonate precipitation. PLoS One 6:e22338

Goldberg WM (1974) Evidence of a sclerotized collagen from the skeleton of a gorgonian coral. Comp Biochem Physiol - Part B Biochem 49:525–526

Goss RJ (1992) The evolution of regeneration: Adaptive or inherent? J Theor Biol 159:241–260

Graham JE, van Woesik R (2013) The effects of partial mortality on the fecundity of three common Caribbean corals. Mar Biol 160:2561–2565

Hall VR (1997) Interspecific differences in the regeneration of artificial injuries on scleractinian corals. J Exp Mar Bio Ecol 212:9–23

Hayward DC, Hetherington S, Behm CA, Grasso LC, Forêt S, Miller DJ, Ball EE (2011) Differential gene expression at coral settlement and metamorphosis - A subtractive hybridization study. PLoS One 6:e26411

Helman Y, Natale F, Sherrell RM, LaVigne M, Starovoytov V, Gorbunov MY, Falkowski PG (2008) Extracellular matrix production and calcium carbonate precipitation by coral cells in vitro. Proc Natl Acad Sci 105:54–58

Henry LA, Hart M (2005) Regeneration from injury and resource allocation in sponges and corals - A review. Int Rev Hydrobiol 90:125–158

Jatkar A (2008) Mucus layer properties and dynamics in reef corals. Ph.D thesis, Newcastle University

Jatkar AA, Brown BE, Bythell JC, Guppy R, Morris NJ (2010) Coral Mucus: The properties of its constituent mucins. Biomacromolecules 11:883–888

Kitchen SA, Crowder CM, Poole AZ, Weis VM, Meyer E (2015) De novo assembly and characterization of four Anthozoan (phylum Cnidaria) transcriptomes. G3 5:2441–2452

Kramarsky-Winter E, Loya Y (2000) Tissue regeneration in the coral Fungia granulosa: The effect of extrinsic and intrinsic factors. Mar Biol 137:867–873

Lesser MP (2004) Experimental biology of coral reef ecosystems. J Exp Mar Bio Ecol 300:217–252

Lesser MP (2006) Oxidative stress in marine environments: Biochemistry and physiological ecology. Annu Rev Physiol 68:253–278

Lesser MP, Shick JM (1989) Effects of irradiance and ultraviolet radiation on photoadaptation in the zooxanthellae of *Aiptasia pallida*: primary production, photoinhibition, and enzymic defenses against oxygen toxicity. Mar Biol 102:243–255

Marshall PA (2000) Skeletal damage in reef corals: relating resistance to colony morphology. Mar Ecol Prog Ser 200:177–189

Mayfield AB, Hsiao YY, Fan TY, Chen CS, Gates RD (2010) Evaluating the temporal stability of stress-activated protein kinase and cytoskeleton gene expression in the Pacific reef corals *Pocillopora damicornis* and *Seriatopora hystrix*. J Exp Mar Bio Ecol 395:215–222

Mayfield AB, Wang LH, Tang PC, Fan TY, Hsiao YY, Tsai CL, Chen CS (2011) Assessing the impacts of experimentally elevated temperature on the biological composition and molecular chaperone gene expression of a reef coral. PLoS One 6:e26529

Meesters EH, Bak RP (1993) Effects of coral bleaching on tissue regeneration potential and colony survival. 96:189–198

Meesters EH, Noordeloos M, Bak RPM (1994) Damage and regeneration: Links to growth in the reef-building coral *Montastrea annularis*. Mar Ecol Prog Ser 112:119–128

Meesters EH, Wesseling I, Bak RPM (1996) Reef-building corals and the relation with colony morphology. Bull Mar Sci 58:838–852

Meesters EH, Pauchli W, Bak RPM (1997) Predicting regeneration of physical damage on a reef-building coral by regeneration capacity and lesion shape. Mar Ecol Prog Ser 146:91–99

Meikle P, Richards GN, Yellowlees D (1988) Structural investigations on the mucus from six species of coral. Mar Biol 99:187–193

Miller DJ, Ball EE, Forêt S, Satoh N (2011) Coral genomics and transcriptomics - Ushering in a new era in coral biology. J Exp Mar Bio Ecol 408:114–119

Mydlarz LD, Palmer CV (2011) The presence of multiple phenoloxidases in Caribbean reef-building corals. Comp Biochem Physiol – Part A 159:372–378

Nagelkerken I, Bak R (1998) Differential regeneration of artificial lesions among sympatric morphs of the Caribbean corals *Porites astreoides* and *Stephanocoenia michelinii*. Mar Ecol Prog Ser 171:279–283

Novak ML, Bryer SC, Cheng M, Nguyen M-H, Conley KL, Cunningham AK, Xue B, Sisson TH, You J-S, Hornberger TA, Koh TJ (2011) Macrophage-specific expression of urokinase-type plasminogen activator promotes skeletal muscle regeneration. J Immunol 187:1448–1457

Palmer CV, Modi CK, Mydlarz LD (2009) Coral fluorescent proteins as antioxidants. PLoS One 4:e7298

Palmer CV, Traylor-Knowles NG, Willis BL, Bythell JC (2011) Corals use similar immune cells and wound-healing processes as those of higher organisms. PLoS One 6:e23992

PEAKS Team (2013) PEAKS 7 User Manual. Waterloo, ON

Polato NR, Voolstra CR, Schnetzer J, DeSalvo MK, Randall CJ, Szmant AM, Medina M, Baums IB (2010) Location-specific responses to thermal stress in larvae of the reef-building coral *Montastraea faveolata*. PLoS One 5:e11221

R Core Team (2017) R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria. URL https://www.R-project.org/

Renegar D-EA (2015) Histology and ultrastructure of *Montastraea cavernosa* and *Porites astreoides* during regeneration and recruitment: Anthropogenic stressors and transplant success. Ph.D. thesis, Nova State University

Ritchie KB (2006) Regulation of microbial populations by coral surface mucus and mucus-associated bacteria. Mar Ecol Prog Ser 322:1–14

Robbart ML, Peckol P, Scordilis SP, Curran HA, Brown-Saracino J (2004) Population recovery and differential heat shock protein expression for the corals *Agaricia agaricites* and *A. tenuifolia* in Belize. Mar Ecol Prog Ser 283:151–160

Rodríguez-Martínez RE, Jordan-Garza AG, Jordan-Dahlgren E (2016) Low regeneration of lesions produced by coring in Orbicella faveolata. Peer J 4:e1596

Rodríguez-Villalobos JC, Work TM, Calderon-Aguilera LE (2016) Wound repair in *Pocillopora*. J Invertebr Pathol 139:1–5

Rossi S, Snyder MJ (2001) Competition for space among sessile marine invertebrates: changes in HSP70 expression in two pacific Cnidarians. Biol Bull 201:385–393

Roth MS, Latz MI, Goericke R, Deheyn DD (2010) Green fluorescent protein regulation in the coral *Acropora yongei* during photoacclimation. J Exp Biol 213:3644–3655

Rougée LRA, Richmond RH, Collier AC (2014) Natural variations in xenobiotic-metabolizing enzymes: Developing tools for coral monitoring. Coral Reefs 33:523–535

Sabine AM, Smith TB, Williams DE, Brandt ME (2015) Environmental conditions influence tissue regeneration rates in scleractinian corals. Mar Poll Bull 95:253–264

Sabourault C, Ganot P, Deleury E, Allemand D, Furla P (2009) Comprehensive EST analysis of the symbiotic sea anemone, *Anemonia viridis*. BMC Genomics 10:333

Seveso D, Montano S, Strona G, Orlandi I, Vai M, Galli P (2012) Up-regulation of Hsp60 in response to skeleton eroding band disease but not by algal overgrowth in the scleractinian coral *Acropora muricata*. Mar Environ Res 78:34–39

Shnit-Orland M, Kushmaro A (2009) Coral mucus-associated bacteria: A possible first line of defense. FEMS Microbiol Ecol 67:371–380

Siboni N, Abrego D, Motti CA, Tebben J, Harder T (2014) Gene expression patterns during the early stages of chemically induced larval metamorphosis and settlement of the coral *Acropora millepora*. PLoS One 9:e91082

Slattery M, Ankisetty S, Corrales J, Marsh-Hunkin KE, Gochfeld DJ, Willett KL, Rimoldi JM (2012) Marine proteomics: A critical assessment of an emerging technology. J Nat Prod 75:1833–1837

Soderhall K, Smith VJ (eds) (1986) Hemocytic and humoral immunity in arthropods. Wiley, New York, pp 251–286

Squires DF (1958) Stony corals from the vicinity of Bimini, Bahamas, British West Indies. Bull Am Museum Nat Hist 115:215–262

Stenken JA, Poschenrieder AJ (2015) Bioanalytical chemistry of cytokines-a review. Anal Chim Acta 853:95–115

Stevens FC (1982) Calmodulin: an introduction. Can J Biochem Cell Biol 61:906–910

Sunagawa S, DeSalvo MK, Voolstra CR, Reyes-Bermudez A, Medina M (2009a) Identification and gene expression analysis of a taxonomically restricted cysteine-rich protein family in reef-building corals. PLoS One 4:e4865

Sunagawa S, Wilson EC, Thaler M, Smith ML, Caruso C, Pringle JR, Weis VM, Medina M, Schwarz JA (2009b) Generation and analysis of transcriptomic resources for a model system on the rise: the sea anemone *Aiptasia pallida* and its dinoflagellate endosymbiont. BMC Genomics 10:258

Swulius MT, Waxham MN (2008) Ca2+/calmodulin-dependent protein kinases. Cell Mol Life Sci 65:2637–2657

Technau U, Steele RE (2011) Evolutionary crossroads in developmental biology: Cnidaria. Development 138:1447–1458

Teitelbaum SL (2000) Bone resorption by osteoclasts. Science 289:1504–1508

Thompson A, Schäfer JJ, Kuhn K, Kienle S, Schwarz J, Schmidt GG, Neumann T, Hamon C (2003) Tandem mass tags: a novel quantification strategy for comparative analysis of complex protein mixtures by MS/MS. Anal Chem 75:1895–1904

Turk V, Stoka V, Vasiljeva O, Renko M, Sun T, Turk B, Turk D (2012) Cysteine cathepsins: From structure, function and regulation to new frontiers. Biochim Biophys Acta - Proteins Proteomics 1824:68–88

Van Veghel MLJ, Bak RPM (1994) Reproductive characteristics of the polymorphic Caribbean reef building coral *Montastrea annularis* III. Fecundity and colony structure. Mar Ecol Ser 109:221–227

Van Oppen MJH, Gates RD (2006) Conservation genetics and the resilience of reef-building corals. Mol Ecol 15:3863–3883

van De Water JAJM, Ainsworth TD, Leggat W, Bourne DG, Willis BL, Van Oppen MJH (2015a) The coral immune response facilitates protection against microbes during tissue regeneration. Mol Ecol 24:3390–3404

van de Water JAJM, Leggat W, Bourne DG, van Oppen MJH, Willis BL, Ainsworth TD (2015b) Elevated seawater temperatures have a limited impact on the coral immune response following physical damage. Hydrobiologia 759:201–214

Voolstra CR, Sunagawa S, Matz MV, Bayer T, Aranda M, Buschiazzo E, DeSalvo MK, Lindquist E, Szmant AM, Coffroth MA, Medina M (2011) Rapid evolution of coral proteins responsible for interaction with the environment. PLoS One 6:e20392

Watanabe Y, Tateno H, Nakamura-Tsuruta S, Kominami J, Hirabayashi J, Nakamura O, Watanabe T, Kamiya H, Naganuma T, Ogawa T, Naudé RJ, Muramoto K (2009) The function of rhamnose-binding lectin in innate immunity by restricted binding to Gb3. Dev Comp Immunol 33:187–197

Wenger Y, Buzgariu W, Reiter S, Galliot B (2014) Injury-induced immune responses in *Hydra*. Semin Immunol 26:277–294

Whitten MMA, Tew IF, Lee BL, Ratcliffe NA (2004) A novel role for an insect apolipoprotein (Apolipophorin III) in β-1,3-glucan pattern recognition and cellular encapsulation reactions. J Immunol 172:2177–2185

Wickham H (2016) Elegant graphics for data analysis. Springer-Verlag, New York

Wootton EC, Pipe RK (2003) Structural and functional characterisation of the blood cells of the bivalve mollusc, *Scrobicularia plana*. Fish Shellfish Immunol 15:249–262

Work TM, Aeby GS (2010) Wound repair in *Montipora capitata*. J Invert Pathol 105:116–119

Young JAC (1974) The nature of tissue regeneration after wounding in the sea anemone *Callactis parasitica* (Couch). J Mar Biol Assoc UK 54:599–617

Zhang X, Guo L, Collage RD, Stripay JL, Tsung A, Lee JS, Rosengart MR (2011) Calcium/calmodulin-dependent protein kinase (CaMK) Iα mediates the macrophage inflammatory response to sepsis. J Leukoc Biol 90:249–261

Zhou Z, Yu X, Tang J, Zhu Y, Chen G, Guo L, Huang B (2017) Dual recognition activity of a rhamnose-binding lectin to pathogenic bacteria and zooxanthellae in stony coral *Pocillopora damicornis*. Dev Comp Immunol 70:88–93